Peasant Resistance in Times of Economic Affluence: Lessons from Paraguay

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Abstract

An influential literature suggests that low agricultural prices prompt peasant revolts in the rural countryside because they reduce returns to agricultural labor and undermine informal subsistence arrangements. We argue that in agrarian frontiers—economic regions with low potential for agroindustrial production and pervasive small-scale farming—positive price shocks drive peasant conflict. High prices encourage large commercial farmers and agribusiness to encroach on the lands of peasants in these areas. Resistance to land encroachment is greater in peripheral areas where organizational legacies and extensive subsistence agriculture allow peasants to pool symbolic and material resources to resist the expansion of capital-intensive commercial agriculture. We provide evidence of this contentious dynamic by using unique municipal-level data from Paraguay. Peasant resistance increases in areas less suitable to commercial agriculture when the prices of agricultural commodities grow over the 2000-2013 period, especially in areas with subsistence settlements and local peasant organizations.

Key words: conflict, rebellion, resource extraction, trade, political economy

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Extant views of rural unrest argue that peasants protest when facing economic distress due to low agricultural prices (Scott, 1976; Wolf, 1969). Steady declines in the price of cash crops decrease the returns to agricultural labor and the incomes peasants earn from cultivation, thus provoking resistance against landowners and state agents who extract a surplus from peasant households (see also Dube and Vargas, 2013). In light of this influential scholarship, the surge of peasant contention in recent decades, a period of extraordinarily high agricultural prices, presents a puzzle. Since the early 2000s, peasants in Argentina (Lapegna, 2016), Mozambique (Clements and Fernandes, 2013), Paraguay (Hetherington, 2011), or Indonesia (McCarthy and Cramb, 2009) have mobilized to repeal the expansion of capital-intensive farming. Historical studies focusing on Latin America describe a similar process of peasants resisting the advent of commercial agriculture during the commodities boom of the late nineteenth century (LeGrand, 1986; Saffon, 2021). In both cases, the expansion of commercial agriculture, motored by rising prices, was the driving force of peasant upheaval in the developing world.

In this article, we argue that economic geography (e.g., Beramendi, 2012) can explain variations in peasant contentiousness during a commodities boom. While low prices decrease the opportunity cost of engaging in contention by jeopardizing incomes, high prices also endanger peasants' livelihoods because they offer incentives to landowners to encroach on their lands.[1] Thus, the impact of prices on contention is not unambiguous as the scholarship suggests; instead, peasant contention varies with exogenous fluctuations in commodity prices and the economic geographic characteristics that enable commercial agriculture, particularly with the suitability of the land for producing valuable cash crops for trade.

Our argument distinguishes between central areas and agrarian frontiers. Central areas are those already incorporated into commercial agriculture, and are usually characterized by higher levels of land suitability to agroindustrial production, closeness to markets, and well-defined ownership rights. In these areas, peasants are tied to the land through tenancy contracts, like sharecropping or rental, with landowners. The agrarian frontier, by contrast, is characterized

[1] Land encroachment is the forceful removal of peasants or indigenous peoples from the land they occupy. This encompasses evictions carried out by thugs or policemen, the destruction of homes and crops, or the harmful utilization of herbicides and pesticides.

by lands that have not been incorporated into commercial agriculture due to lower suitability, remoteness, and unclear property rights. As commodity prices rise, these lands become commercially attractive. Hence, high commodity prices spur land encroachment and expose peasants on the agrarian frontier to the expansion of capital-intensive farming.

When confronted with land encroachment, peasants can either abandon their land—with or without compensation—or resist. Resistance requires symbolic and material resources to generate collective action in response to grievances. These resources often rely on prior experiences of peasant contention and organization (Lapegna, 2016; Ondetti, 2008). Moreover, the communal traditions of peasant settlements practicing subsistence agriculture promote social norms of exchange and reciprocity that facilitate the coordination of contentious defensive behavior (Scott, 1976; Thompson, 1971). We therefore expect peasant resistance to be greater in regions of the agrarian frontier where peasants have been organized and subsistence agriculture is practiced.

We apply this framework to explain peasant resistance to land encroachment in Paraguay during the commodities boom of the early twenty-first century. Paraguay is largely a rural country. As of 2013, more than forty percent of the population was still living in rural areas. During the commodities boom, Paraguay became a top producer of capital-intensive agricultural commodities, especially soybeans. Agroindustrial production, which was typical in the central belts surrounding the capital city of Asunción, expanded eastward into the country's agrarian frontier, the Eastern Region, where peasant colonies had settled in the 1960s and 1970s. Previous studies have linked the 2000s price hikes to land encroachment, the displacement of peasant settlements, and a contracted demand for rural labor (Riquelme and Vera, 2013).

Empirically, we test our argument using a unique dataset of rural unrest in Paraguay at the municipal level based on newspaper archives spanning 2000-2013, a time of peaking yet fluctuating agricultural prices. Our dataset contains rich information about the key dimensions of contention, including the actors involved as well as their demands and protest repertoires. We combine our measures of contention with satellite, census, and administrative data. We proxy economic geography by using data on land suitability to Paraguay's most exported agricultural commodities, as determined by agroclimatic factors. Whereas central lands are more suitable

to commercial agriculture and have been integrated into world markets at early stages, frontier lands have not been incorporated into commercial production and are devoted to small-scale farming.

We evaluate our argument by examining, quantitatively, the interaction between exogenous shifts in the international price of Paraguay's main commodities and land suitability at the municipal level. We show that higher prices heighten the intensity of peasant resistance, especially in municipalities where land suitability is low—i.e., the less fertile frontier municipalities. Furthermore, we show that this differential effect is heightened in municipalities where local peasant organizations and subsistence agriculture are prevalent. Our results are robust to alternative econometric specifications and placebo tests. We contextualize our econometric findings using in-depth interviews with stakeholders as well as qualitative data from our original database on rural unrest.[2]

Our work contributes to different strands of research. First, we dispute an earlier literature suggesting that higher agricultural prices benefit peasants by boosting their income in the context of free trade (Bates, 1981). We show that, in contexts of urbanizing societies and capitalized agricultural sectors which reduce demands for rural labor, higher prices foster land encroachment and force peasants to choose between displacement or contention. In this sense, our argument is related to the expansion of so-called "flex-crops" such as soybeans, maize, or sugar that triggered processes of land grabbing and peasant expulsion (Borras Jr. et al., 2012).

Our findings also complement a literature linking booming prices of capital-intensive commodities with rural armed conflict (Dal Bó and Dal Bó, 2011). This literature focuses exclusively on lootable hydrocarbons or minerals that are ecologically concentrated in deposits, which attract rebel peasants motivated by greed. By contrast, we show that higher prices of capital-intensive agricultural commodities—a primary activity that has been ignored in the political economy literature—can also ignite various forms of peasant contentiousness (not necessarily armed responses) through grievances, not greed.

Furthermore, we conceive of peasant resistance as a reactive form of contention that aims

[2]Our qualitative research comprises 35 semi-structured interviews in the cities of Asunción, Lambaré, Luque, and San Lorenzo conducted in between 2014-2020.

to defend established claims (Tilly, 1978). In our case, it emerges when peasants' hold on land is threatened. However, peasant studies have overwhelmingly focused on armed group action—such as guerrilla insurrection or revolution—while ignoring the moderate varieties of peasant mobilization we study in this article, including roadblocks, sit-ins at public places, or demonstrations (Trejo, 2016).

Finally, we build on the historically-based scholarship on peasant revolts at the turn of the twentieth century, when commodity prices also spiked. In particular, Saffon (2021) suggests that the intensity and scope of peasant mobilization varied according to the preexisting land tenure "ecosystem"—different configurations of property rights and community-level resources. Our characterization of the agrarian frontier resembles her "*colono* frontier ecosystem." But whereas she focuses on how peasant uproar surged in response to land dispossessions, we are more interested in how the interplay between commodity prices and economic geography shapes peasant resistance.

Commodity Prices, Geography, and Contention

The literature on political economy has emphasized the difficulties for peasant collective action based on spatial dispersion, scarce resources, and embeddedness in subordinated relations of subsistence (Bates, 1981). A longstanding consensus maintains that the introduction of commercial agriculture into peasant communities, who live on the verge of subsistence, has been a major cause of peasant resistance (Scott, 1976; Thompson, 1971; Wolf, 1969).[3] Commercial agriculture exposes peasants to market insecurities—as wages and the value of crop yields fluctuate with international prices—and makes them vulnerable to macroeconomic shocks.

Classic agrarian studies, such as Scott (1976) or Wolf (1969), suggested that steep declines in grain prices drive peasant contentiousness.[4] Economic busts (determined by recessions or

[3]We conceive peasant resistance as a collective overt response rather than "everyday acts of resistance" (Scott, 1976).

[4]We refer to "peasants" as those populations who work in farms in exchange for wages or independently cultivate a parcel for subsistence purposes and/or selling a surplus of their harvest in domestic markets. We use the term "landowners" as a synonym of landed elites, denoting large and mid-sized commercial farmers as well as

climatic disasters) beget deprivation among peasants because landowners and state officials shift the burden to them by demanding greater exactions in the form of cash rents or levies. Steadily low prices shatter their livelihoods and leave them no choice other than rebellion. More recently, scholars in political economy have underlined the opportunity cost of rural conflict (Dal Bó and Dal Bó, 2011; Dube and Vargas, 2013; Guardado, 2018; Hidalgo et al., 2010). A negative shock to commodity prices lowers the earnings peasants make from cultivation, thus reducing the opportunity cost of engaging in contention. In contrast, when prices increase, incomes improve and the opportunity cost increases, thereby hindering peasants' incentives to revolt.

The waves of peasant mobilization that ensued throughout the 2000s commodity windfall are puzzling in light of this literature. Why do peasants participate in protests in a commodities boom, when they are supposedly benefiting from price upswings? And why do they protest in some places but not others?

To answer these questions, we differentiate between two types of economic geography.[5] On the one hand, central areas involve traditional belts of large-scale commercial agriculture that were formed during the initial phases of state formation. Cities and infrastructure for trade, such as roads, railroads, and ports, were strategically located when these regions were included in world trade corridors facilitating integration into the export economy (Engerman and Sokoloff, 2012). Examples are the Argentine Pampas, the Brazilian South, or the Corn Belt in the Midwestern United States. Central lands feature high levels of land suitability to agroindustrial production, clearer property rights, and accessibility and proximity to agricultural markets. In those areas, peasants are usually wage laborers working in commercial estates and cultivating a parcel of land through tenancy agreements with landowners, like sharecropping or rental.

On the other hand, agrarian frontiers are peripheral areas that have not yet been incorporated into commercial agriculture. These are vast hinterlands informally occupied by peasants for subsistence of smallholding agriculture. Frontiers share common characteristics like low

agribusiness firms.

[5]By economic geography, we refer to cross-regional differences in the conditions that enable commercial production (see Beramendi, 2012).

land suitability to commercial production, remoteness from cities and ports, and ill-defined landownership such as squatting or communal rights (Alston et al., 1999; LeGrand, 1986). Examples are the Brazilian Amazon, the Argentine Chaco, or the Paraguayan Eastern Region.

In much of the developing world, particularly in Latin America, the peasantry comprised a class of frontier settlers (i.e., *colonos*) who colonized wastelands by means of slash-and-burn agriculture, either spontaneously or through state colonization schemes (Alston et al., 1999; LeGrand, 1986; Saffon, 2021). Because their lands are readily available for cultivation and are not incorporated into commercial agriculture, these peasants are especially exposed to land encroachment in a commodities boom. Rising agricultural prices provide unusual opportunities for profit and encourage landowners, who produce cash crops for trade, to expand agroindustrial production in central areas to agrarian frontiers. Even though peripheral lands are less suitable, soaring prices makes these lands attractive and incentivize landowners to enter the agrarian frontier and encroach on tracts possessed by peasants using violent evictions, legal fraud, or both.[6] Borras Jr. et al. (2012) point to agricultural commodities known as "flex crops" as providing strong incentives for land encroachment. These crops involve capitalized agriculture and have interchangeable uses (e.g., foodstuffs, fuels) that increase their commercial value in international markets while having economic and environmental consequences for peasant communities (Hall et al., 2015). Peasant struggles against soybeans, oil palm, or sugarcane in the agrarian frontiers of South America (Grajales, 2015; Lapegna, 2016), Southeast Asia (Hall, 2011; McCarthy and Cramb, 2009), or Africa (Clements and Fernandes, 2013) epitomize peasant resistance in the new millennium.

Hence, our theory stresses that higher agricultural prices—especially, those of "flex crops"—should prompt peasant resistance to land encroachment. Encroachment is more pervasive in the agrarian frontier because these lands have not been incorporated into commercial agriculture; prior efforts by peasant *colonos* may have already cleared the land for agriculture; and property rights are weak. We therefore expect:

[6]Agrarian frontiers that resemble Saffon's "*colono* frontiers," as opposed to forests or swathes of barren land, have already been prepared for agriculture through slash-and-burn techniques, which provides incentives to landowners to encroach them in times of high prices.

Hypotheses 1. *As commodity prices increase, episodes of peasant resistance to land encroachment should occur more frequently on agrarian frontiers—that is, in areas less suitable to commercial agriculture.*

Instead of resisting encroachment, it is often easier for peasants to acquiesce to landowners' claims and strike individual deals that do not require coordination with other peasants (Bates, 1981). For example, peasants can abandon or sell their lands (at prices below the market value) and move to urban outskirts to work in the informal economy (Grajales, 2015; Lapegna, 2016). This has been commonplace in the twenty-first century because capital-intensive agriculture reduces labor demand, whereas in the past commercial expansion forced landowners into rental or sharecropping arrangements with peasants.

To mount defensive collective action against land encroachment, peasants require organizational resources. We follow a large literature on peasant contentiousness that highlights the importance of peasant organizational legacies and the type of agriculture peasants practice. First, past experiences of peasant organizing can provide organizational skills, frames, and support structures for contentious politics. In much of 1950s and 1960s Latin America, Communist intellectuals and Catholic priests (especially, those advocating Liberation Theology) helped peasants to organize in peasant leagues (Brockett, 1991; Zamosc, 1986). These leagues rebelled against landowners who forced subsistence peasants to cultivate cash crops for trade—instead of letting them produce food for their families—in exploitative working conditions. They demanded land reform, and some called for armed struggle. Although military governments effectively suppressed peasant leagues, readiness for mobilization in the wake of new grievances under democratic rule was one of the principal legacies of the peasant leagues of the past in the agrarian frontiers of Argentina, Brazil, or Paraguay (Lapegna, 2016; Ondetti, 2008; Riquelme, 2003). We thus suggest the following hypothesis:

Hypotheses 2. *As commodity prices increase, episodes of peasant resistance to land encroachment should occur more frequently on agrarian frontiers, especially in locations where peasants are organized.*

A second mechanism refers to the social organization of subsistence agriculture—as

densely-settled settlements characterized by communal traditions of autonomy, solidarity, and reciprocity, which improve peasants' ability to resist the threats of commercial agriculture at the local level (Scott, 1976; Thompson, 1971). These "subsistence ethics" promote community-level exchanges and independence from agricultural markets, thus facilitating contention when peasants' daily lives are in peril. Saffon's (2021) case studies show that the coordination of resistance to dispossession in Mexico and Colombia during the first commodities boom was greater in closely-knit subsistence communities, such as *colono* settlements or indigenous *pueblos*, where members decided community affairs autonomously and pooled resources to resist evictions or make petitions in capital cities. Conversely, and according to this view, peasant households that commercialize cash crops at a small scale (e.g., smallholders or family farms) tend to be more fragmented and less internally able to soften the impact of exogenous market dislocations. Hence, subsistence agriculture's communal traditions of solidarity and reciprocity are crucial to carry out peasant protests and circumvent outmigration. Our third hypothesis is the following:

Hypotheses 3. *As commodity prices increase, episodes of peasant resistance to land encroachment should occur more frequently on agrarian frontiers, especially in locations where subsistence agriculture is practiced.*

Our in-depth interviews indicate that these two mechanisms are complementary and at play when peasants collectively defend their lands from encroachment. In the next section we turn to the case of Paraguay to contextualize our theory and the indicators we use to test our main hypotheses. The content is heavily informed by in-depth, semi-structured interviews with relevant actors as well as the newspaper articles used in the construction of our database.

Commercial Agriculture and the Peasantry in Paraguay

Commercial agriculture has been Paraguay's mainstay. Historically, it was limited to the Central Zone—central and southern rural areas situated in the Central, Cordillera, Guairá, Misiones, Ñeembucú, and Paraguarí departments. These zones are endowed with some of the country's most fertile lands and were appealing for trade because they were near Asunción,

where the country's largest port is located (Zoomers, 1988). Commercial agriculture was dominated by large haciendas and ranches dedicated to cotton farming and cattle, with trading in world markets and clearer property rights than other regions of the country. Peasants in the Central Zone practiced subsistence farming and worked for landowners as sharecroppers in exchange for small pieces of land called *minifundios*.

The rest of Paraguay was not commercially integrated into the country's export agriculture (Nickson, 1981). Outside the Central Zone, land is less suitable for commercial production because of rugged terrain, daunting vegetation, and remoteness from Asunción. This includes the Eastern Region, a vast hinterland in Alto Paraná, Amambay, Caaguazú, Canindeyú, Concepción, and portions of Itapúa and San Pedro. In this region, land was mostly state-owned but idle and sparsely occupied by indigenous populations, a few forestry companies extracting tannin, and Brazilian pioneers.

Rising population density since the 1950s as well as increasing agricultural mechanization exacerbated frictions between commercial farms and *minifundios* over scarce land resources and parcel boundaries. These conflicts led to the formation of the Christian Agrarian Leagues (LAC), a radical agrarian movement made up of communal organizations (Telesca, 2014). With the leadership of Jesuit and Franciscan priests, the LAC sought to combat land tenure injustices through biblical teachings, rural schools, and, occasionally, occupations of unproductive estates. The greater influence of Marxist ideas inspired some of their members to embrace armed conflict as a means to achieve land reform. The military government of Alfredo Stroessner, who ruled on behalf of the Colorado Party (1954-1989), confronted these organizations with a mix of repression and compensation. The government created the Rural Welfare Institute (IBR) in 1963 to resettle peasants in the new peasant colonies of the Eastern Region. By promoting the "March to the East," Stroessner aimed to mitigate class conflict and deactivate peasant demands for land reform in the Central Zone (Nickson, 1981).

Peasants resettled in eastern departments by clearing forests and preparing soils for cultivation. The IBR assigned parcels of approximately 20 hectares to each migrant peasant family, though the average size of the parcels varied by department.[7] According to Rojas and Areco's

[7]It ranged from 15 hectares in San Pedro to 25 hectares in Concepción (Rojas and Areco, 2017).

data (2017), the IBR created 459 peasant colonies between 1963-1989, each controlling, on average, 616 parcels of 18 hectares. Peasant settlers mostly cultivated subsistence crops such as cassava, beans, and peanuts, and raised livestock for their households, and supplied food to local towns (Palau et al., 2007). High prices in the 1970s and low capital inputs stimulated many peasants to commercialize cotton, reducing the farmland dedicated to subsistence farming. The area cultivated with cotton went from 81,000 hectares in 1973 to 312,000 hectares in 1979 (Rojas, 2016). Despite modest income increases, cotton brought many peasant families into indebtedness due to taxation and payments to cotton-ginning factories. Many other peasants, however, continued to produce food for their families (Palau et al., 1986).

The policy of peasant resettlement undermined peasants' property rights over colonized lands (Nickson, 1981; Rojas and Areco, 2017). Land titles granted to peasants were provisional and thus not eligible as collateral. Numerous peasants also migrated spontaneously searching for free abundant lands, overcrowding the IBR's capacity to manage existing colonies and create new ones. Colonized areas lacked roads, technical assistance, and access to public services. Moreover, rampant corruption pushed the IBR to misallocate land grants to cronies of the Colorado Party and foreign investors. As a result, peasant settlers had to rely on squatting and communal forms of land occupation that were not formally recognized. By the time of Paraguay's 1989 democratization, as Hetherington (2011) notes, far-reaching colonization programs had ended and the Eastern Region was now home to masses of peasant cultivators whose tenure conditions remained precarious.

Since then, Paraguayan commercial agriculture has changed in significant ways. The adoption of mechanized equipment during the 1990s increased agricultural productivity and reduced demands for farm labor (Hanratty and Meditz, 1990). An agribusiness mode of commercial production was consolidated. A new class of landowners made of large commercial farmers and agroindustrial groups produced commodities for trade while forging ties with multinational firms to acquire inputs and contract stockpiling, trading, and food-processing services (Rojas, 2016). Soaring international prices of agricultural commodities with potential for multiple uses, combined with technological innovations (e.g., genetically-modified seeds, no-till farming, agrochemicals) in capital-intensive farming, inaugurated an economic bonanza in the new

millennium. Paraguay rose as one of the world's top soybean producers while also increasing the agroindustrial production of maize, sugarcane, and rice (Fogel and Riquelme, 2005).

These exogenous changes generated a land rush toward the Eastern Region among landowners from central areas and Brazilian entrepreneurs—who were attracted by Paraguay's lower taxes and land prices. Because eastern lands had already been cleared by peasant settlers and lacked legally-recognized titleholders, landowners had additional incentives to expand to this region (Galeano, 2012). As a result, commercial agriculture expanded eastward. Soybean monoculture epitomizes this geographic and temporal trend. Soybeans in 1960-1961 were primarily found in the southern department of Paraguarí, with only 1,300 planted hectares (Palau et al., 1986). In 2001-2002, soybeans had penetrated the borderlands of Alto Paraná, Canindeyú, and Itapúa, growing to 1.5 million planted hectares. By 2013, this number more than doubled.[8]

Peasant Resistance to Land Encroachment

The 2000s commodities super cycle made peasants in the agrarian frontier increasingly vulnerable to encroachment (Palau et al., 2007; Riquelme and Vera, 2013). Landowners intruded on peasants' lands using sheer force or by exposing peasant families to herbicides and pesticides meant for genetically-modified seeds. These incursions were backed by the country's two wealthiest associations of commercial farmers and ranchers, the Farmers' Union Syndicate (UGP) and the Paraguayan Rural Association (ARP). Both dismissed peasants as economically backward and incapable of producing at high rates (see Galeano, 2012).

Against this backdrop, peasants had the option to sell or abandon their lands and move to the city, or stay and resist. The Paraguayan literature provides extensive evidence of peasants migrating from rural to urban areas (Fogel and Riquelme, 2005; Palau et al., 2007; Rojas and Areco, 2017). Population censuses also report that interdepartmental migration from the Eastern Region to urban areas increased by 2002 (DGEEC, 2005). While collective resistance to encroachment may have been the exception rather than the rule, we recorded a large number of

[8]CAPECO, `https://capeco.org.py/area-de-siembra-produccion-y-rendimiento/` (accessed on February 25, 2021).

land-related conflicts over this period. Drawing on our press archives on rural conflict and data from FAO's Statistical Database (FAOSTAT), Figure 1 shows the annual variation in events of peasant resistance to encroachment[9] across Paraguay's three regions—i.e., Central, Eastern, and Chaco—and the average annual price in U.S. dollars of the country's most exported cash crops (soybeans, maize, rice, and sugar) between 2000-2013. Although peasant resistance oscillated over the period, it peaked around the years of global price hikes—2004, 2008, and 2012. In these years, the bulk of peasant upheaval, between 57.1 and 100 percent of conflicts, transpired in the Eastern Region.

These episodes of resistance were directly related to the expansion of commercial agriculture. In response, peasants often staged land invasions and invoked the IBR—renamed as the Land and Rural Development National Institute in 2004—to acknowledge their land claims. Peasants clashed both with the police that tried to evict them, and with commercial farmers, as the following episode from our archival database illustrates:

> A dead man and more than 50 people arrested is the result of one of the most violent evictions ever recorded in the San Pedro department. It was yesterday, in the Cuapé ranch, whose lands were occupied by nearly 500 peasants…the police came in the settlement where landless peasants initially tried to confront them with machetes and clubs.[10]

This protest was staged by the MCNOC, a national peasant movement, who blamed the ARP for the police repression.[11] Our records also include many instances of landowners displacing peasants by applying large doses of herbicides and pesticides close to their homes. In these cases, peasants tried to sabotage commercial harvests and block landowners from spraying agrochemicals:

> About 200 peasants from the MCNOC occupied the TZ property, in Toro Pirú, from the Guayabí district, of 2,700 hectares …Peasants made it clear that…they

[9] We detail how the "peasant resistance" variable is constructed in the measurement subsection.

[10] "Los desalojos se cobraron otra vida campesina," *Última Hora*, November 5, 2004.

[11] "MCNOC culpa al fiscal de la muerte de Aureliano Espínola," *Última Hora*, November 5, 2004.

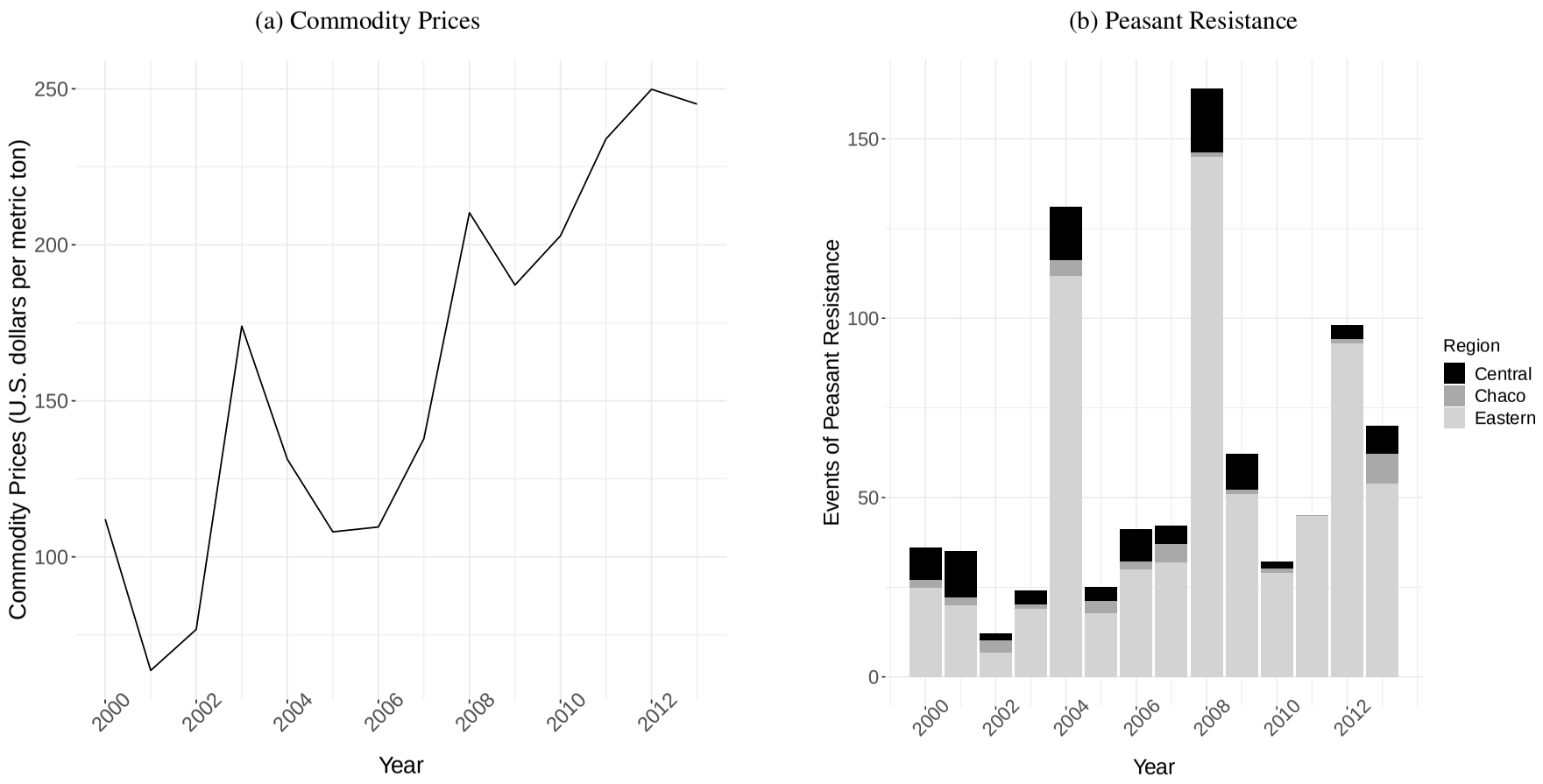


Figure 1: Commodity prices and peasant resistance, 2000-2013

> won't stop fighting for something they consider legitimate: "The fumigation they are carrying out is terrible…the landowner is leasing this property to Brazilian farmers who cultivate soybeans and corn [and] our crops are getting ruined because of the fumigation."[12]

In interviews with peasant organizers, it became clear that preexisting organizational resources within peasant communities helped peasants to coordinate collective responses towards encroachment. The LAC's prior struggles activated feelings of grievance among peasants and provided them with support structures of activists, grassroots movements, and sources of funding for mobilization. As the national secretary of an important peasant movement put it, "[our resistance] is the ash of the LAC, a result of their struggle [and] some of the LAC's principles, and one of those principles is social class consciousness as a peasantry."[13] After the LAC were effectively suppressed in 1976, some of their surviving rank-and-file members regrouped to found local peasant committees that became active in the early days of democratic rule, eventually converging into the formation of national peasant federations (Fogel, 2001; Hetherington, 2011; Riquelme, 2003).[14] These federations provided logistical and financial assistance at the local level during land invasions and roadblocks. They also organized rallies to Asunción in the 1990s and 2000s to demand land titles. Of special salience in the national arena were the FNC, MCP, or MCNOC, which arose as coalitions of local LAC offshoots.[15]

Our interviews also point to the importance of subsistence communities for collective action. A peasant national leader underscored their autonomy from commercial agriculture: "resisting in the countryside means, by definition, producing for feeding ourselves…it can't be done if we are producing at a large scale."[16] According to peasant activists, the implicit rule is to support fellow peasants in distress who are about to lose their land, regardless of any af-

[12] "Campesinos entran a una finca ajena y destruyen el maizal," *Última Hora*, July 19, 2008.

[13] Author interview with P.O., Asunción, August 7, 2014.

[14] Author interview with M.G., Asunción, August 12, 2014.

[15] Author interview with L.A. (Asunción, August 6, 2014), P.O. (Asunción, August 7, 2014), and M.G. (Asunción, August 12, 2014). For a full list of Paraguayan peasant federations as well as local movements refer to Dávalos and Rodríguez (1994).

[16] Author interview with P.O., Asunción, August 7, 2014.

filiation to a specific local committee.[17] This is relevant for land invasions: "peasant fellows from different settlements help each other…during a land invasion."[18] Furthermore, solidarity and reciprocity allow peasants to pool material resources for launching and sustaining protests over time. In times of mobilization, peasants finance themselves in different ways: "by giving away their own produce or selling food in rural fairs and donating a share of that sale, but also by direct monetary contributions to fund rallies to Asunción."[19]

These testimonies are in line with an extensive literature (e.g., Hanratty and Meditz, 1990; Hetherington, 2011; Palau et al., 2007; Rojas and Areco, 2017) showing that communal values of solidarity, reciprocity, and autonomy have traditionally permeated Paraguayan subsistence communities.

Empirical Strategy

To test the effect of commodity prices on peasant resistance in agrarian frontiers, we compiled a dataset that includes Paraguay's 223 municipalities.[20] The data span the years 2000-2013, a period marked by a global rise in commodity prices. Our unit of analysis is the municipality-year and the number of observations is 3,122.

Measurement

Dependent Variable. Our main dependent variable is the annual number of events of peasant resistance to land encroachment. These are occurrences in which peasants engage in contentious collective action against landowners—large commercial farmers and agribusiness firms—or the police to fight back, or protest against, attempts at seizing their land. We measured this variable by comprehensively coding press archives from *Última Hora*, one of Paraguay's top-selling newspapers. This is the only nationwide newspaper during our period of

[17] Author interview with M.G., Asunción, August 12, 2014.

[18] Author interview with E.F., Lambaré, March 30, 2015.

[19] Author interview with M.G., Asunción, August 12, 2014.

[20] Because the number of municipalities changed during our period of analysis, we map all our variables to the 223 municipalities that existed in 2002. We excluded Asunción because it is a fully urbanized district.

study whose archives were available and complete.[21] We recorded the municipality where the conflict occurred, the actors involved, the issue under dispute, the protest repertoires, and the organization of the involved actors, if any. We distinguished issues between access to land, agricultural regulations (e.g., credit, prices, transportation costs), the environment, and repression. Repertoires that peasants carry out include roadblocks, sit-ins at public places, land invasions, public demonstrations, or the destruction of government or private property.

Our dependent variable includes conflicts over access to land, environmental conflicts over deforestation or agrochemicals that displace peasants from their lands, and protests against violent evictions. We focus on land-related conflicts involving peasants or indigenous peoples, on the one hand, and landowners or state agents, on the other hand. These are occurrences in which peasant or indigenous groups are the aggrieved party, whereas landowners or the state are the aggressors. We excluded conflicts between peasants and indigenous peoples (e.g., over demarcation of land boundaries) and between landowners and the state (e.g., over export taxes). We identified unique instances of peasant resistance, which required linking multiple stories reporting the same event and disaggregating events occurring in several places but reported simultaneously. Our final dataset includes 817 distinct events of peasant resistance. The number of conflicts at the municipality-year level ranges from 0 to 15, with a mean of 0.26 and a standard deviation of 1.04. We normalize this variable using a logarithmic transformation.[22]

The dependent variable only includes instances of peasant resistance reported by a national newspaper, thus raising potential measurement concerns. For instance, peasant federations may increase reporting by making certain conflicts salient at the national level. Our measure may be contaminated by other factors, too, including the commercial value of land or whether the newspaper has a correspondent in the locality. To account for these (observed and unobserved) factors that may determine whether events of rural conflict are reported in the data, most of our models include departmental or municipal fixed effects. To further check the robustness of our main results against the systematic underreporting or overreporting of events, we re-run our

[21] The archives of *Última Hora* were retrieved from its repositories and the National Library of Paraguay, both located in Asunción.

[22] Let y be the number of conflicts in a given year, our dependent variable is transformed using the following formula: $\log(y + \sqrt{y^2 + 1})$.

main models using (i) a binary dependent variable indicating whether an instance of peasant resistance occurred in a municipality-year or not and (ii) restricting our sample to municipalities that have reported at least one event of peasant resistance over the period (see Albertus et al., 2016; Hidalgo et al., 2010).[23] These approaches are less sensitive to measurement error.

Explanatory Variables. To examine how the interplay between commodity prices and economic geography shapes peasant resistance, we collected data on Paraguay's main "flex crops:" maize, rice, soybeans, and sugar. They are Paraguay's most internationally-traded agricultural commodities, accounting for 30 percent of the country's total exports in 2014.[24]

Our first explanatory variable is the log of the average producer price of Paraguay's main agricultural commodities, in U.S. dollars per metric ton, as reported by FAOSTAT. Commodity prices in Paraguay are exogenous. Although Paraguay is a top producer of some of these commodities, it falls far behind Brazil, the United States, or Argentina. Paraguay is a price-taker in international agricultural markets, having no capacity to alter its supply. For example, the country is the fourth global producer of soybeans, but only contributes three percent of the total output.[25]

Second, we leverage variation between central and frontier areas by examining satellite data on land suitability to commercial agriculture, as determined by agroclimatic factors such as climate, soil nutrients, and terrain ruggedness over the 1960-1991 period. The data were taken from FAO's Global Agro-Ecological Zones (GAEZ). Suitability is measured as the log of the average potential yield for Paraguay's main agricultural commodities, in metric tons per hectare. We computed the municipality-level average of land suitability and then divided it by the national average suitability.[26]

We identify frontier municipalities as those with less suitable soil for producing cash crops

[23] Appendix Tables A5 and A6.

[24] OEC, `https://oec.world/` (accessed on January 26, 2021).

[25] FAOSTAT, `http://www.fao.org/faostat/en/#data/QC/` (accessed on January 26, 2021).

[26] We downloaded GAEZ satellite rasters on suitability for intermediate inputs, both irrigated and rain-fed, and spatially merged them with a layer of 2002 municipal boundaries to obtain measures of municipality-level suitability for each of the four crops. This gives us an average value of suitability within each municipality's polygon, weighted by the area of overlap with each suitability grid cell.

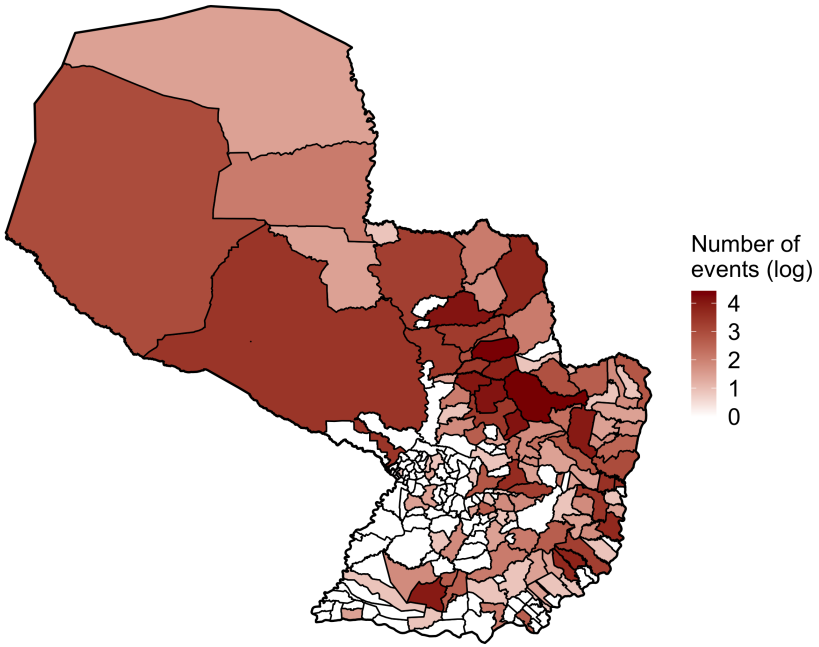
for trade. This measure is a better proxy for economic geography than actual production—e.g., if landowners are already producing in the municipality. First, land suitability closely conforms to our theoretical mechanism: we are interested in how landowners' incentives to encroach on peripheral lands—which are less fertile and have not yet been incorporated into commercial agriculture—trigger peasant resistance, not on whether landowners are able to expand production. Second, the decision to plant any of these crops may be endogenous to contention. All else being equal, landowners should avoid municipalities where peasants can engage in collective resistance. Furthermore, planting decisions may be correlated with unobserved or hard-to-measure factors, including ties between landowners and local political elites that may impact the likelihood of rural conflict. By contrast, land suitability is time-invariant and exogenous to local political dynamics, thus mitigating concerns about reverse causation and confounding.[27]

Figure 2 presents the geographic distribution of peasant resistance in 2003-2013 in panel (a), next to the level of land suitability across Paraguayan municipalities in panel (b). Comparing both maps, we can see that peasant resistance and land suitability are inversely correlated, in line with our theoretical expectations. The cluster of municipalities in the Eastern Region, which show high or moderately high levels of peasant unrest, also exhibit the lowest levels of land suitability to commercial agriculture. Conversely, the municipalities of the southern and southeastern belts—the Central Zone nearby Asunción—show low or nonexistent levels of peasant resistance, as well as the highest potential for commercial production.

We examine the importance of organizational resources for contentious politics using two sources of data. First, we rely on satellite data on the number of subsistence settlements in 1990-1992. These communities are dense, vaguely-demarcated areas of human settlement, also dubbed *núcleos*, in which subsistence agriculture is practiced. Guyra Paraguay, an environmental NGO, provided the data. Second, we gathered data on the territorial presence of local peasant committees from the 1992-1993 guide prepared by Dávalos and Rodríguez (1994). Peasant committees are the lowest tier of national peasant federations, with affiliated members

[27]Data on cultivation at the municipal level in Paraguay are only available for 1991 and 2008, when national agricultural censuses were conducted. In Appendix Table A14, we show that the expansion of commercial agriculture at the municipal level between these years is negatively correlated with land suitability.

(a) Peasant Resistance to Land Encroachment



(b) Land Suitability to Commercial Agriculture

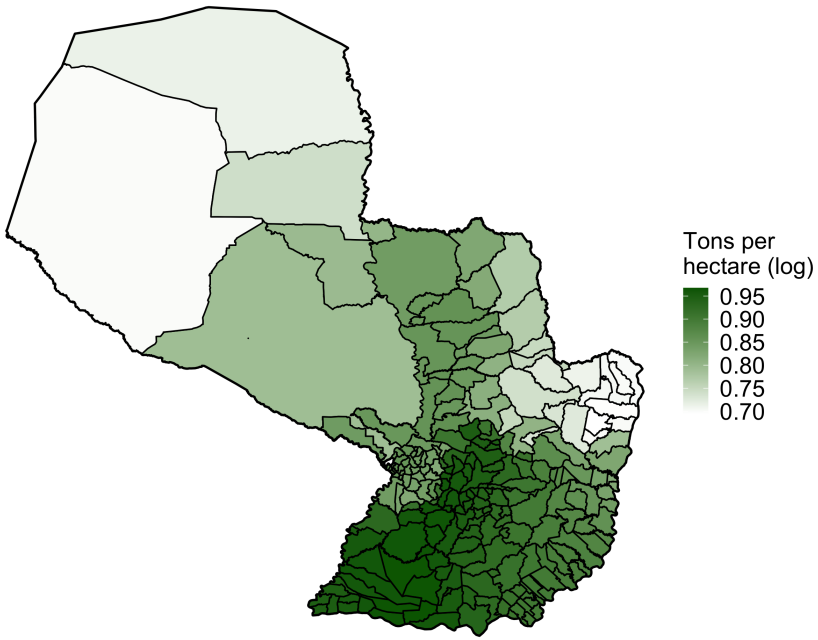


Figure 2: Peasant resistance (2003-2013) and land suitability by municipality

operating within the community or neighborhood. This guide, while obviously outdated, captures well the legacies of peasant organizing dating from the 1960s LAC that resurfaced during the democratic transition (1989-1992). These new organizations mobilized landless peasants to demand titling, access to public services, and agricultural credit from the state (see Riquelme, 2003).

We rely on organizational legacies as we do not know whether these peasant organizations survived, or if new ones were created into the 2000s. However, we know that peasant organizations were not formed in response to new grievances resulting from the 2000s commodities boom, as they predate price hikes.

Control Variables. We include two control variables whose omissions may confound our results. First, weak property rights in land, which are typical in agrarian frontiers, are measured as the share of a municipality's occupied farmland that is not legally titled—tenants are squatters or do not enjoy legal rights. We expect this variable to have a positive effect on peasant resistance, as occupants lacking formal property rights are more vulnerable to encroachment. The data come from the 1991 agricultural census. Second, distance to markets is measured as a municipality's minimum distance, in kilometers, to any of the eight cities that are trading hubs, where agricultural commodities are exported. Distance should be positively correlated with peasant resistance, because more remote lands have not yet been commercially integrated. Both untitled farmland and distance to markets are time invariant. In all our models, we also control for the log of the municipal population. The data for trading hubs and population come from Paraguay's census bureau (DGEEC).

Estimation

We focus on two main empirical strategies. Our first model estimates changes in events of peasant resistance as a function of changes in commodity prices and levels of land suitability:

$$y_{it} = \beta_1 P_t + \beta_2 S_i + \beta_3 (P_t \times S_i) + \mathbf{X}_{it} + \delta_k + \gamma_t + \varepsilon_{it} \tag{1}$$

where i, k, and t index each municipality, department, and time period, respectively; y_{it} is the

log number of events of peasant resistance in a given municipality i and year t; P_t is the log of commodity prices at year t; S_i is the log of the municipal-level measure of land suitability; $\mathbf{X}_{it}$ is a matrix of controls, as defined above; and δ_k and γ_t are departmental and year dummies, respectively. Time-period effects control for yearly shocks common to all municipalities. We opt for dummies at the department (not municipal) level in our main specification because fixed-effects estimators rely on within-unit variance, and several explanatory variables, including land suitability, are time invariant. Since the interaction between our key independent variables ($P_t \times S_i$) does vary within units, we also present models with municipal fixed effects. The estimand β_3 captures the differential effect of commodity prices on peasant resistance at different levels of land suitability. Our main hypothesis is that a rise in commodity prices causes an increase in peasant resistance to land encroachment over less suitable lands ($\beta_3 < 0$).

The main identification assumption is $E(\varepsilon_{it}|P_t, S_i, \mathbf{X}_{it}, \delta_k, \gamma_t) = 0$. In this setup, serially correlated errors may arise because of time-varying omitted variables or misspecified persistence in the dependent variable. To deal with these potential misspecifications, we first estimate Equation 1 using both fixed-effects and two-way random-effects regression models. While the fixed-effects model solves the problem of omitted variables with constant effects, the random-effects model constrains the sample-to-sample variability of estimates. In both cases, we cluster standard errors by municipality to address heteroskedasticity and serial correlation.[28]

Our second strategy relies on directly modeling the dynamic structure of the data-generating process by fitting a linear model of the following form:

$$y_{it} = \beta_1 P_t + \beta_2 S_i + \beta_3 (P_t \times S_i) + \mathbf{X}_{it} + \sum_{j=1}^{p} \sigma_j y_{it-j} + \gamma_t + \varepsilon_{it} \tag{2}$$

This equation adds an arbitrary number of lags of the dependent variable to deal with potential time-varying unobservables that might bias our estimates. We estimate this model using OLS and report panel-corrected standard errors. Adding lags of the dependent variable con-

[28]Our results are also robust to a hybrid, fixed-effects negative binomial estimator in which our dependent variable is measured as event counts (Appendix Table A7). We choose least squares as the primary estimator as there are both excess zeros as well as overdispersion in our data, making it problematic to fit count models with fixed effects (Allison and Waterman, 2002).

trols for a number of potential time-varying confounders, including any effects that past protest episodes may have had on current episodes of peasant resistance.[29]

To evaluate the interaction effect between prices and land suitability, next to our linear models we present results from the binning model proposed by Hainmueller et al. (2019). This model estimates the moderating effect of a given variable by splitting the sample into roughly equal-sized groups.[30] We analyze the data at three levels of the suitability variable: low (first tercile), medium (second tercile), and high (third tercile). The model includes department and year fixed effects and the same battery of controls used in the main models. We evaluate the conditional impact of subsistence agriculture and peasant organization by estimating Equation 1 in separate samples: one for municipalities whose number of subsistence settlements and committees are equal or less than the median, and a second one, for municipalities with values greater than the median. Kam and Franzese Jr. (2007) show that split-sample analyses ease the interpretation of conditional effects and are practical for exploring complex heterogeneous relationships.[31]

Results

Table 1 presents our main results. The linear models in the table estimate the interaction effect of commodity prices and land suitability on peasant resistance. Column 1 presents estimates for the differential effect of price increases on peasant resistance in municipalities that are less suitable to commercial agriculture, controlling only for department- and year-fixed effects. Column 2 adds control variables to the model. Columns 3-4 report results from two-way random-effects regressions, with and without controls. Columns 5-6 add three lags of the dependent variable. Finally, column 7 presents a model with municipal-level dummies. Prices and time-invariant

[29]Because prices are exogenous to municipal-level characteristics, and suitability is time-invariant, we are not concerned with past outcomes affecting these variables.

[30]This procedure limits concerns about common support and nonlinearity in the data.

[31]Three-way interactions require more demanding assumptions and are especially problematic to estimate (Hainmueller et al., 2019; Kam and Franzese Jr., 2007). Our results are almost identical when using a three-way interaction in the full sample (Appendix Table A8) and the number of subsistence hectares and peasants affiliated to committees as moderators (Appendix Table A9).

regressors are excluded from this model.

Table 1 provides supporting evidence for Hypothesis 1. The coefficients for commodity prices are positive across all models, indicating that when the international prices of agricultural commodities increase, peasant resistance to land encroachment increases, too. This positive effect is decreasing in land suitability. Prices have a differentially positive effect on resistance in the less suitable municipalities—i.e., agrarian frontiers. This finding is consistent and statistically significant across our seven models. The coefficient for distance to trading hubs is also positive and statistically significant, providing additional support to our argument. During a period of generally high prices, resistance concentrates in more remote municipalities. Surprisingly, the coefficient for untitled farmlands changes its sign across models and is not significant, though this could be explained by Paraguay's absence of a national cadastral survey mapping landed properties accurately (Fogel and Riquelme, 2005).

We illustrate the moderating effect of land suitability in Figure 3, which shows the linear marginal-effect estimate from the fixed-effects model in column 2, Table 1, and the coefficient estimates from the binning model. Comparing both models we can conclude that the linearity assumption is supported by the data (see also Appendix Figure A6). The magnitude of the price effect decreases as land suitability increases. In other words, the positive effect of prices on peasant resistance is concentrated in the municipalities with less fertile lands—in the Eastern Region. According to the binning estimates, moving one standard deviation upward in commodity prices increases resistance in municipalities in the lowest tercile of suitability by 6 percentage points. By contrast, at medium and maximum levels of suitability—i.e., the Central Zone—a rise in prices barely affects levels of resistance. For the binning estimates, the effect of prices on peasant resistance fails the conventional statistical significance test. These differences in marginal effects among municipalities in frontier and central areas are statistically significant at conventional levels (see Appendix Table A2).

Turning to Hypothesis 2 and 3, we investigate the differential effect of price increases on resistance, contingent on LAC organizational legacies and subsistence farming. We test the incidence of these variables by dividing the sample into municipalities above and below, or equal to, the median number of peasant committees and subsistence settlements. Table 2 re-

Table 1: Peasant Resistance, Commodity Prices, and Land Suitability, 2000-2013

	Model 1	**Model 2**	**Model 3**	**Model 4**	**Model 5**	**Model 6**	**Model 7**
Commodity prices (log)	1.382^{***}	1.331^{***}	1.395^{***}	1.396^{***}	0.923^{**}	1.046^{**}	
	(0.367)	(0.389)	(0.399)	(0.421)	(0.463)	(0.474)	
Land suitability (log)	7.144^{***}	7.893^{***}	7.142^{***}	7.807^{***}	4.989^{*}	5.976^{**}	
	(2.163)	(2.332)	(2.069)	(2.270)	(2.696)	(2.776)	
Commodity prices × Land suitability	-1.492^{***}	-1.491^{***}	-1.492^{***}	-1.513^{***}	-1.010^{**}	-1.150^{**}	-1.492^{***}
	(0.395)	(0.422)	(0.393)	(0.419)	(0.473)	(0.488)	(0.408)
Distance		0.002^{***}		0.001^{***}		0.001^{***}	
		(0.001)		(0.0004)		(0.0002)	
Untitled farmland		0.0003		−0.001		−0.0003	
		(0.002)		(0.002)		(0.001)	
Population (log)		0.149^{***}		0.114^{***}		0.062^{***}	
		(0.021)		(0.021)		(0.017)	
Resistance$_{t-1}$ (log)					0.257^{**}	0.234^{**}	
					(0.106)	(0.103)	
Resistance$_{t-2}$ (log)					0.147	0.109	
					(0.114)	(0.110)	
Resistance$_{t-3}$ (log)					0.184	0.146	
					(0.118)	(0.114)	
Department FE	Yes	Yes	No	No	Yes	Yes	No
Municipality FE	No	No	No	No	No	No	Yes
Year FE	Yes	Yes	No	No	Yes	Yes	Yes
N	3122	2884	3122	2884	3122	2884	3122

Note: Standard errors clustered by municipality (Models 1-4 and 7) and panel corrected (Models 5-6) in parentheses. Constants estimated but not reported. The unit of analysis is the municipality-year. Models 1-2 include department fixed effects. Models 3-4 include two-way random effects. Models 5-6 include lagged terms of the dependent variable. Model 7 include municipal fixed effects.
$^{*}p < .1$, $^{**}p < .05$, $^{***}p < .01$

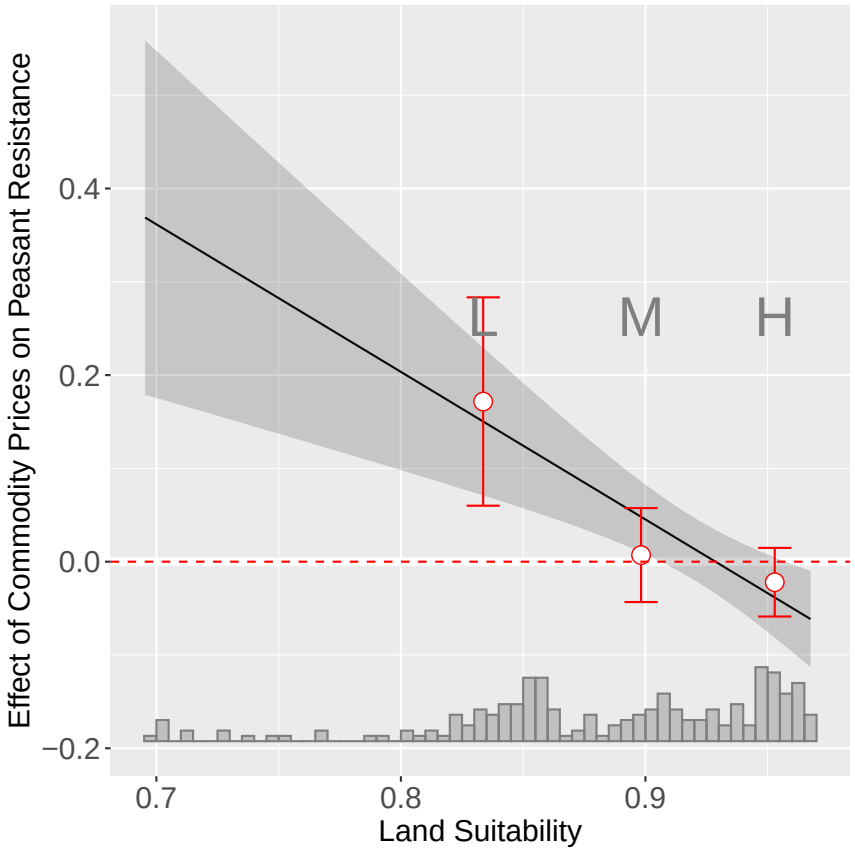


Figure 3: Marginal effect of commodity prices on peasant resistance by land suitability. The histogram represents the distribution of municipalities at different levels of land suitability. Grey bands represent 95% confidence intervals. L, M, and H represent the 95% intervals for the marginal effects at the low, medium, and high terciles.

ports our fixed-effects model with and without controls for each subsample. Consistent with our hypotheses, there is no evidence of an effect of prices on resistance, regardless of land suitability, for municipalities with few or no subsistence settlements or committees. By contrast, prices have a positive, large, and statistically significant effect on resistance in less suitable municipalities with strong LAC legacies and prevalent subsistence farming.

Figure 4 illustrates these differences graphically. Panel (a) investigates the impact of prices across levels of land suitability for municipalities with low or no subsistence farming. Panel (b), in turn, presents results for municipalities with high subsistence farming. For municipalities with low prevalence of subsistence farming, higher commodity prices do not affect peasant resistance. This is true across the whole range of land suitability. By contrast, higher prices increase peasant resistance in municipalities where subsistence settlements predominate. The effect of commodity prices on resistance is decreasing in levels of land suitability in these municipalities. Increasing commodity prices by one standard deviation in municipalities with the least suitable soils increases peasant conflict in these districts by 12 percentage points—i.e., twice the effect reported for the full sample. For municipalities with medium and high levels of land suitability, the effect of prices is precisely estimated near zero and fails to achieve statistical significance. The difference in the marginal effects for municipalities with prevalent subsistence agriculture between frontier and central areas is statistically significant at conventional levels.

We find a similar relationship when examining the role of peasant organizational legacies, presented in panels (c) and (d) in Figure 4. We find that prices do not increase peasant resistance in municipalities without peasant committees. If anything, higher prices have a negative, yet small, effect on resistance in municipalities characterized by high levels of land suitability and where committees are absent. In contrast, prices increase resistance in municipalities with committees and characterized by low or medium levels of land suitability. If commodities prices increased by one standard deviation, peasant resistance would increase by roughly 12% in these municipalities. In turn, there is no effect of prices on resistance in municipalities with the highest levels of suitability and with a strong organizational legacies.[32]

[32]See Appendix Tables A3 and A4 for full results.

Table 2: Peasant Resistance, Subsistence Agriculture, and Peasant Organization, 2000-2013

	$\leq med$(Settlement)		$> med$(Settlement)		$\leq med$(Committee)		$> med$(Committee)	
	Model 1	**Model 2**	**Model 3**	**Model 4**	**Model 5**	**Model 6**	**Model 7**	**Model 8**
Commodity prices (log)	0.631	0.202	2.274***	2.508***	0.614	0.568	2.805***	2.205***
	(0.531)	(0.501)	(0.500)	(0.534)	(0.451)	(0.455)	(0.715)	(0.668)
Land suitability (log)	2.484	1.551	13.089***	14.472***	4.001	4.402	15.060***	12.644***
	(2.955)	(2.959)	(3.148)	(3.185)	(2.726)	(2.698)	(4.247)	(4.012)
Commodity prices × Land suitability	−0.726	−0.336	−2.396***	−2.680***	−0.729	−0.695	−2.985***	−2.396***
	(0.560)	(0.526)	(0.545)	(0.580)	(0.468)	(0.471)	(0.791)	(0.749)
Distance		0.002**		0.003***		−0.001*		0.004***
		(0.001)		(0.001)		(0.0003)		(0.001)
Untitled farmland		−0.002		−0.00001		−0.0004		0.0003
		(0.002)		(0.002)		(0.001)		(0.003)
Population (log)		0.142***		0.127***		0.064***		0.205***
		(0.028)		(0.028)		(0.013)		(0.035)
Department FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	1582	1414	1540	1470	1638	1596	1372	1288

Note: Standard errors clustered by municipality in parentheses. The unit of analysis is the municipality-year. Models 1-2 and 5-6 are estimated on municipalities whose levels of subsistence settlements and peasant committees are equal or less than the median. Models 3-4 and 7-8 are estimated on municipalities whose levels of subsistence settlements and peasant committes are greater than the median.
$^{*}p < .1$, $^{**}p < .05$, $^{***}p < .01$

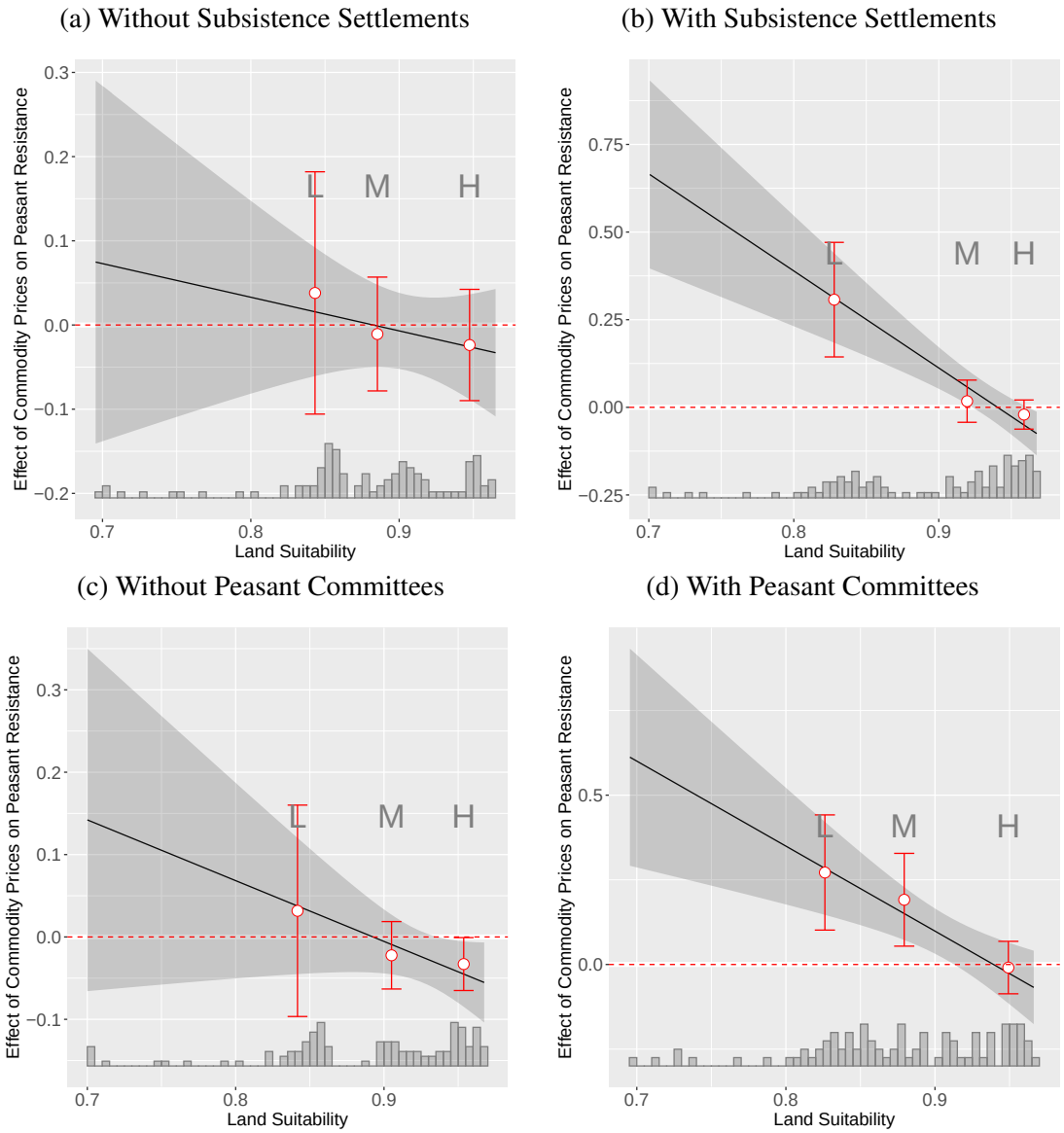


Figure 4: Marginal effect of commodity prices on peasant resistance by land suitability for municipalities with and without subsistence agriculture and organizational legacies. The histograms represent the distribution of observations at different levels of land suitability. Grey bands represent 95% confidence intervals. L, M, and H represent the 95% intervals for the marginal effects at the low, medium, and high terciles.

In short, our results show support for our hypotheses about the effect that the expansion of commercial agriculture has had on peasant resistance to land encroachment in Paraguay. When commodity prices grow, so does resistance in the less suitable frontiers where land suitability to commercial production is low (Hypothesis 1). That effect is heightened in areas with organizational legacies (Hypothesis 2) and where subsistence farming is practiced (Hypothesis 3), further supporting our theory.

Placebo Tests and Alternative Explanations

We conducted placebo tests to lend credibility to our main results. One concern is the type of rural conflict. We theorize that, during a commodities boom, peasant conflict should be more over land and less over income—the latter, as the conventional literature suggests, is common in periods of steadily low prices that threaten peasant earnings. If this is true, we should not observe effects similar to those reported in Table 1 for income-related peasant protests. We address this question by re-running our main models using episodes of peasant resistance to income shortages as the dependent variable. We coded from our archival database all the instances of rural conflict in which income is the grievance—e.g., peasants or indigenous peoples protesting against low wages or lack of credit.

The results are presented in Appendix Table A10 and Figure A7. The coefficient for commodity prices has the opposite sign and is not statistically significant. The figure shows that this effect is slightly decreasing on land suitability in the linear model, and statistically significant for values roughly between the low and high terciles. Yet, the binning estimator shows that this effect is nonlinear and nonmonotonic, being indistinguishable from zero for medium and high levels of suitability. Therefore, prices do not have a differentially positive effect on income-related conflict in less suitable areas.

Another concern is that subsistence settlements might be simply picking up the presence of peasants instead of capturing subsistence agriculture's resources for contention. We deal with this issue by distinguishing between municipalities that cultivate more subsistence crops relative to cotton. Whereas some peasants switched to cotton in the 1970s because it required low capital inputs and prices were high, their credit needs—profits were used to pay cotton-grinning

factories, suppliers, and traders—coupled with taxes on cotton exports resulted in unsustainable debts. Since commercial production atomizes peasants and makes them susceptible to market disruptions (e.g., Scott, 1976), increased reliance on small-scale cotton in relation to subsistence agriculture should hinder peasants' ability to resist. In our interviews, peasant leaders stressed the incompatibility of cotton commercialization with subsistence farming and the detrimental consequences that cotton has had for peasant collective action.[33]

We measure the trade-off between subsistence and cotton farming using a ratio of the number of hectares planted with subsistence crops to those with cotton, for farms of 20 hectares or less. The subsistence-cotton ratio proxies for whether peasants in a given municipality lean toward subsistence or commercial agriculture. We use a dummy indicating whether the subsistence-cotton ratio is greater than the sample median. The data for planted cotton was taken from the 1991 agricultural census. The results are presented in Appendix Table A11 and Figure A8. The marginal effect of prices for low levels of suitability is indistinguishable from zero for municipalities where cotton dominates over subsistence crops, as shown in panel (a). Conversely, a positive and significant effect holds for municipalities where subsistence agriculture is large relative to cotton, as shown in panel (b). Thus, small-scale commercial agriculture exerts a demobilizing effect on peasants.

Finally, it could be that the collective action capacity of landowners to penetrate the frontier and despoil peasants of their lands—and not peasant organizational legacies—is what catalyzes resistance (see Albertus et al., 2016). Peasants might be carrying out defensive actions only in places where landowners are well-organized and capable of threatening their communities. The anecdotal evidence exhibited in the case section suggests that Paraguayan landowner associations are sometimes behind encroachments. We explore this possibility by using data on the local societies, chapters, and cooperatives belonging to the ARP and UGP, Paraguay's largest landowner associations. The data come from their lists of affiliates and district offices.

Appendix Table A12 and Figure A9 show that there is no significant difference between municipalities with and without landowner associations—the coefficients for the interactions between commodity prices and land suitability in both samples are large, statistically signifi-

[33]Author interview with P.O., Asunción, August 7, 2014.

cant, and with the predicted signs. Therefore, landowner associations do not seem to have a differential impact on peasant resistance.

Conclusions

The commodities super cycle of the new millennium was a time of unprecedented economic growth for developing countries, only comparable to the 1890s and the post-WWII era (Erten and Ocampo, 2013). Paraguay benefited exceptionally from these booming years. Between 2000-2013, the Paraguayan economy grew, on average, 4.5% annually.[34] The agricultural sector was the main driver of this growth: as a share of total exports, agricultural exports rose from 40 to 70 percent during this period.[35]

In this context, heightened peasant unrest is puzzling. Prior studies argued that the intrusion of commercial agriculture into peasant lives fuels contention, with falling prices operating as the key mechanism. When prices plummet and grain markets crash, incomes shrivel and the peasantry's communal security mechanisms are pulverized, thus stimulating resistance against landowners and the state. Conversely, price upswings were expected to reward peasants by raising rural wages and allowing them to reap the benefits of trade, thereby increasing the opportunity cost of engaging in contentious behavior. Hearkening back to Scott (1976) or Wolf (1969), we maintain that the commercialization of the countryside drives peasant resistance. By distinguishing between different economic geographies, nevertheless, we also maintain that extraordinary periods of economic affluence can engender turmoil between landless peasant cultivators and landowners seeking to expand commercial production to new frontiers. Thus, the relationship between price cycles and peasant contentiousness is not unambiguous as previously suggested in the literature.

In Paraguay, peasants with ill-defined property rights inhabiting the Eastern Region's less suitable and more remote lands were put in a defensive position in years of high commodity prices. Commercial farmers and agribusiness firms with linkages to multinational corpora-

[34]IMF, `https://www.imf.org/en/Data` (accessed on February 2, 2021).

[35]OEC, `https://oec.world/en/profile/country/pry` (accessed on January 26, 2021).

tions encroached on frontier lands and forced peasants to sell or abandon their homes and migrate to urban areas. When peasants could pull resources together, they resisted by clashing with the police, invading neighboring lands, blockading roads, or staging other forms of contentious collective action. We further identified two possible sources of these organizational resources, which were often mentioned in our interviews. In municipalities where Christian peasant leagues had mobilized in the past, peasants could draw on their organizational legacies to fight land encroachment. Similarly, peasant communities involved in subsistence farming could rely on local traditions of solidarity and reciprocity to collectively resist land grabbing.

Our theory on positive price shocks, economic geography, and peasant organizational resources is not unique to Paraguay. Church and Maoist leadership in the 1950s northeastern leagues stirred landless peasants in Brazil's Amazon basin to resist the expansion of soybean fields from core belts (Ondetti, 2008). Our theory can be generalized outside Latin America as well. A case in point is Mozambique (see Clements and Fernandes, 2013), where the peripheral tropical savannas have been the target of agroindustrial soybean and sugarcane projects with ties to foreign private banks. The subsistence customs of peasant communities as well as the local organizations of the UNAC—a peasant front built on the struggles of anti-Portuguese independence movements—helped peasants to contest these projects.

Future studies should disentangle the effects of price fluctuations for different agricultural commodities—e.g., crops that are more or less intensive in capital and require biotechnology or agrochemicals. The literature investigating peasant contentiousness has assumed that commercial agriculture is a labor-intensive activity. Higher agricultural prices increase rural wages and offer labor opportunities to the rural poor. However, in the last three decades, the production of commodities for trade (particularly, of "flex crops") has become a highly capitalized activity. Even though developing countries may benefit from better terms of trade and greater fiscal surpluses for subsidizing urban constituencies, the peasant and indigenous populations living in the peripheries—whose economic survival hinges on having land for cultivation—might be hurt. Capital-intensive farming engenders land deprivation, depletes forests, and causes environmental damages. In that regard, our paper suggests a more nuanced view of the winners and losers of the 2000s commodities boom.

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